

Computational Modeling of Bound States in the Yukawa Potential

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Abstract

The Yukawa potential is a short-range interaction proposed by Hideki Yukawa in 1935 to describe the strong nuclear force through the proposal of a massive particle, later identified as the meson. His paper [3] explains how the meson mediates the attractive force between nucleons and overcomes the repulsive electromagnetic force at short distances, contributing to the stability of atomic nuclei. This model acts as a screened Coulomb potential, with a similar $1/r$ dependence and acting over distances on the order of 10^{-15} m (1 fm), but differing from the Coulomb potential by the presence of an exponential decay factor. We aim to build a computational solver exploring bound states by solving the non-relativistic radial Schrödinger equation. The main focus is to solve for low-energy bound state eigenenergies and eigenfunctions using numerical methods and to plot the corresponding wavefunctions as functions of radial distance r , but also examines how bound states depend on the strength and range of the potential and to identify the conditions under which bound states emerge or disappear.

References

- [1] D. J. Griffiths, *Introduction to Quantum Mechanics*, 2nd ed. (Pearson Prentice Hall, 2005).
- [2] J. J. Wang, *Computational Modeling and Visualization of Physical Systems with Python*, (Wiley, 2016), Chap. 9; Sec. 5, Apps. A, C.
- [3] H. Yukawa, “On the Interaction of Elementary Particles,” *Proc. Phys.-Math. Soc. Japan* **17**, 48–57 (1935).